# E-Commerce Conversion Prediction - Methodology Report

Summer Analytics 2026 | Week 2 Mini-Hackathon | IIT Guwahati CAC

## 1. Problem Statement

Binary classification task: predict whether an e-commerce user converts (1) or not (0), evaluated on F1 Score. Dataset contains 13,000 labelled records (train + public_test) and 3,000 unlabelled private_test records across 12 input features covering user demographics, browsing behaviour, traffic source, device, and marketing context.

## 2. Exploratory Data Analysis

Class distribution is imbalanced at 30.5% positive (converted). Missing values were present in Age (11.4%), Income (7.6%), and Time_On_Site (14.2%). Correlation analysis revealed **Pages_Viewed** (r=0.308) and **Products_Viewed** (r=0.306) as the strongest predictors. Discount_Seen (r=0.112) and Previous_Purchases (r=0.105) provided secondary signal. Browser_Version, Campaign_Code, and Age showed near-zero raw correlation with the target.

## 3. Data Preprocessing

Key steps applied consistently across train and test:

- Merged train.csv and public_test.csv (labeled) into a 13,000-row training set.
- Label-encoded Device_Type and Traffic_Source after fitting on all data to prevent unseen labels.
- Imputed missing values using column-wise **median** (robust to outliers); fitted on train only.
- No standard scaling applied - tree-based models are scale-invariant.

## 4. Feature Engineering

19 new features were derived from the original 12, totalling 31 features:

| Feature Group | Features Created | Rationale |
|---|---|---|
| Engagement Interactions | Pages_x_Products, Pages_sq, Products_sq, Total_Engag | ementCapture non-linear browsing depth |
| Discount Amplification | Discount_x_Pages, Discount_x_Products, Discount_x_Prev | Discount effectiveness by engagement leve |
| Time Depth | Time_per_Page, Time_per_Product | Session quality indicators |
| Loyalty Flags | Loyal_User (>2 purchases), New_User (0 purchases) | Purchase history segmentation |
| Channel Flags | Is_Paid, Is_Email, Is_Social, Is_Mobile, Is_Desktop | One-hot of top channels |
| Threshold Flags | High_Pages (>=10), High_Products (>=5) | Binary engagement thresholds |
| Geographic | Tier1_High_Income | High-value city-income interaction |

## 5. Model Development & Ensemble

Three tree-based classifiers were selected for their suitability with tabular data and class imbalance. Class imbalance was addressed via **scale_pos_weight** (XGBoost), **is_unbalance=True** (LightGBM), and **class_weight='balanced'** (RandomForest). Final predictions used a weighted soft-vote ensemble:

| Model | 5-Fold CV F1 (mean ± std) | Weight in Ensemble |
|---|---|---|
| XGBoost (n=600, depth=6, lr=0.04) | 0.5236 ± 0.0121 | 35% |
| LightGBM (n=600, depth=6, lr=0.04) | 0.5296 ± 0.0060 | 35% |
| RandomForest (n=400, depth=10) | 0.5471 ± 0.0098 | 30% |
| **Ensemble (OOF threshold-tuned)** | **0.5577** | ---- |

## 6. Threshold Optimisation

Default classification threshold of 0.5 is suboptimal for imbalanced datasets. Out-of-fold (OOF) ensemble probabilities were used to sweep thresholds from 0.30 to 0.70 (step 0.01) and select the value maximising F1 on the training distribution. The optimal threshold was **0.40**, reflecting the class imbalance and improving recall on the minority (converted) class. Final private test prediction distribution: 1,596 converted (53.2%) and 1,404 not converted (46.8%).

## 7. Summary

The pipeline combines labeled public_test data into training, engineers 19 interaction features focused on browsing depth and discount engagement, and applies a three-model soft-vote ensemble with threshold tuning - achieving an OOF F1 of 0.5577. All steps are reproducible in notebook.ipynb.